

HERWIJNEN, Netherlands

WMO: 063560

Lat: 51.8581N

Lon: 5.1452E

Elev: 1

StdP: 101.31

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-6.7	-4.5	-9.7	1.6	-5.0	-7.2	2.1	-4.0	14.3	9.6	12.9	9.0	2.7	80	0.565

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.2	28.5	20.3	26.3	19.4	24.5	18.4	21.3	26.8	20.2	25.0	19.2	23.2	3.3	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.3	14.1	23.6	18.4	13.3	22.3	17.6	12.6	21.3	61.7	26.9	58.0	25.2	54.7	23.4	25.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	11.2	9.6	8.4	DB	-9.9	32.8	4.7	1.7	-13.3	34.0	-16.1	35.0	-18.7	36.0	-22.2	37.3
(5)				WB	-10.3	23.9	4.6	1.1	-13.6	24.7	-16.3	25.3	-18.9	26.0	-22.3	26.8

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.5	3.3	3.9	6.3	9.6	13.3	16.1	18.0	17.5	14.9	11.1	7.0	4.0
	DBStd	6.28	4.06	3.85	3.40	3.63	3.53	3.07	3.02	2.61	2.76	3.29	3.39	4.21
	HDD10.0	876	209	172	121	52	9	0	0	0	1	26	99	187
	HDD18.3	2954	467	404	374	263	159	81	44	47	108	224	340	443
	CDD10.0	1041	1	1	6	38	112	183	247	233	147	61	10	2
	CDD18.3	77	0	0	0	1	4	13	33	21	4	0	0	0
	CDH23.3	675	0	0	0	8	49	135	276	166	40	2	0	0
	CDH26.7	164	0	0	0	0	6	33	78	41	7	0	0	0
Wind	WSAvg	4.0	4.8	4.7	4.4	3.8	3.7	3.5	3.5	3.2	3.3	3.9	4.2	4.7
Precipitation	PrecAvg	839	70	64	53	45	62	71	86	82	70	74	80	81
	PrecMax	1039	127	129	98	99	130	131	178	188	174	185	136	164
	PrecMin	590	5	16	12	0	21	16	33	14	10	22	11	21
	PrecStd	117	34	29	28	24	29	33	41	46	38	35	31	38
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.8	14.1	18.2	24.6	27.4	30.3	32.1	30.7	27.2	22.0	16.4	13.4
		MCWB	11.3	10.5	12.9	15.8	19.2	21.2	21.2	21.6	19.7	17.4	14.1	11.8
	2%	DB	11.2	11.7	15.4	21.1	24.5	26.9	28.6	27.4	24.2	18.8	14.5	12.2
		MCWB	10.0	9.6	11.2	14.4	16.7	19.3	20.3	20.3	18.8	16.0	12.9	11.0
	5%	DB	10.0	10.5	13.2	18.4	22.5	24.4	26.2	24.9	21.9	17.1	13.2	11.1
		MCWB	8.8	8.9	10.2	13.1	15.9	18.3	19.5	19.2	17.6	14.8	11.8	9.9
	10%	DB	8.7	9.2	11.5	16.0	20.1	22.3	24.2	22.8	19.8	15.9	11.9	9.8
		MCWB	7.7	7.9	9.3	11.8	14.9	17.2	18.7	18.0	16.3	14.0	10.7	8.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.5	11.2	13.3	17.0	20.1	22.2	22.7	22.6	20.7	17.9	14.6	12.2
		MCDB	12.4	12.8	16.6	23.5	26.1	28.4	28.7	28.4	25.8	20.8	15.6	13.1
	2%	WB	10.2	10.0	11.9	14.9	18.2	20.3	21.3	21.0	19.3	16.3	13.1	11.2
		MCDB	11.2	11.3	14.6	20.0	22.6	25.3	27.1	26.0	23.3	18.5	14.3	12.1
	5%	WB	8.9	9.0	10.7	13.6	16.6	19.0	20.2	19.8	18.1	15.3	11.9	10.1
		MCDB	9.8	10.2	12.7	17.8	21.0	23.3	25.1	23.9	21.0	17.0	13.0	11.0
	10%	WB	7.8	8.0	9.6	12.3	15.3	17.8	19.1	18.7	17.0	14.2	10.8	8.8
		MCDB	8.7	9.2	11.4	15.4	19.3	21.6	23.1	21.8	19.5	15.6	11.8	9.8
Mean Daily Temperature Range		MDBR	4.7	5.9	7.8	9.8	9.7	9.6	9.2	9.3	8.8	7.1	5.5	4.6
	5% DB	MCDBR	5.0	7.1	11.0	13.6	13.2	13.3	13.1	12.7	11.7	8.2	6.1	4.7
		MCWBR	4.6	5.5	7.5	7.9	6.9	7.1	6.2	6.6	6.7	5.6	5.1	4.3
		MCDBR	5.1	6.3	9.7	12.2	11.7	11.7	11.7	11.5	10.4	7.9	6.0	4.9
		MCWBR	4.7	5.1	7.0	7.5	6.7	6.8	6.1	6.5	6.3	5.8	5.2	4.5
Clear-Sky Solar Irradiance	taub	0.344	0.367	0.401	0.427	0.434	0.429	0.439	0.429	0.403	0.380	0.356	0.339	
	taud	2.299	2.269	2.184	2.157	2.187	2.220	2.201	2.249	2.307	2.334	2.311	2.284	
	Ebn at Noon	640	734	778	806	820	827	810	797	773	710	616	572	
	Edh at Noon	67	90	117	135	137	134	135	122	103	83	65	58	
All-Sky Solar Radiation	RadAvg	0.72	1.41	2.60	4.18	5.06	5.43	5.12	4.33	3.16	1.83	0.84	0.55	
	RadStd	0.08	0.27	0.34	0.43	0.38	0.40	0.58	0.37	0.28	0.15	0.08	0.07	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (88 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page